

Dalan

A Smart Route Recommendation and
Departure Optimization Mobile Application

for Private Vehicle Drivers in Cebu Province

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Capstone Proposal

Statement of the Problem

Private vehicle owners in Cebu Province face daily transportation challenges that drive up costs, waste fuel, and harm the environment.

1



Traffic Congestion

Longer travel times and delayed arrivals

2



Rising Fuel Costs

Increasing transportation expenses daily

3



No Cost Estimates

Cannot estimate fuel cost before a trip

4



Carbon Emissions

Unaware of emissions from route choices

5



Road Hazards

Flooding, closures, and poor air quality

Existing Solutions & Gaps

Google Maps



- Route guidance and navigation
- Estimated travel times
- Real-time traffic conditions
- Alternative route suggestions

Waze



- Community-driven traffic reporting
- Real-time navigation
- Alternative route suggestions
- Incident and road condition reports

Key Gaps in Current Solutions

✗ No fuel cost estimates (PHP)

✗ No carbon emission calculations

✗ No departure timing optimization

✗ No Cebu-specific hazard information

✗ No route scoring by cost & sustainability

✗ No eco-score or green route incentives

Introducing

Dalan

A mobile application designed to help private vehicle owners in Cebu Province make smarter travel decisions by evaluating routes across multiple dimensions.

Evaluates routes based on:

Travel Time

Fuel Cost (₱)

Traffic Conditions

Fuel Consumption

CO₂ Emissions

Local Hazards



Efficiency · Affordability · Sustainability

Core Features

Part 1 of 2

01



Smart Route Recommendation

Compare multiple routes showing estimated travel time, fuel consumption (km/L), fuel cost in ₱, and CO₂ emissions. Prioritize by Fastest, Cheapest, or Greenest.

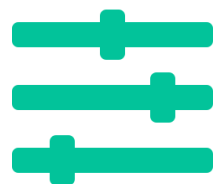
02



Departure Optimization

Before navigation, DALAN analyzes congestion and recommends delaying departure if significant traffic is detected, showing potential time and fuel savings.

03



Dynamic Route Adjustment

While navigating, the app continuously monitors traffic. When unexpected congestion appears, it calculates and recommends an alternate route in real-time.

Core Features

Part 2 of 2

04



Personalized Vehicle Profiles

Users input fuel type and engine displacement during onboarding. DALAN uses this to generate accurate, personalized fuel consumption, cost, and emission calculations.

05



DOE Live Fuel Price Integration

Integrates weekly fuel prices from the Department of Energy Philippines. Calculations automatically update every week to reflect current pump prices per fuel type.

06



Eco Score Dashboard

After each trip, DALAN generates an Eco Score tracking total CO₂ saved, fuel cost savings, and green route selections to encourage sustainable travel habits.

07



Cebu-Specific Hazard Layer

Provides Cebu-relevant hazard data: flood-prone road segments, air quality advisories, and event-related closures that are not available in standard navigation apps.

What Makes DALAN Different?

Feature	Google Maps	Waze	DALAN
Real-Time Navigation	✓	✓	✓
Traffic Monitoring	✓	✓	✓
Alternative Routes	✓	✓	✓
Fuel Cost Estimation (₱)	✗	✗	✓
Carbon Emission Calculation	✗	✗	✓
Departure Optimization	✗	✗	✓
Eco Score Dashboard	✗	✗	✓
Cebu-Specific Hazard Layer	✗	Limited	✓

Expected Benefits



Economic

Reduced fuel expenses

Better route selection

More efficient travel planning



Environmental

Lower carbon emissions

Increased environmental awareness

Promotion of sustainable transport



Safety

Hazard awareness before travel

Improved route planning

Better real-time decisions

DALAN drives smarter, more cost-efficient, and environmentally responsible travel in Cebu.

Conclusion

Current navigation applications focus primarily on speed and traffic. DALAN goes further.

1

Fuel Cost Awareness

Real ₱ estimates before every trip

2

Carbon Emission Analysis

Route scoring that includes sustainability

3

Departure Optimization

Wait-or-go guidance to avoid congestion

4

Personalized Profiles

Accurate data based on your vehicle

5

Cebu Hazard Information

Local flood, air quality, and closure alerts



DALAN

Smarter · Affordable
Sustainable

Thank You

Questions and Discussion

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